

**Photon energy should not be equated to kinetic energy of the electron.*

- 4 (a)** The number of nuclei inside the box does not change! This is as when the cobalt-60 decays, another nucleus will be formed and as such, the number of nuclei in the box stays constant. The appropriate definition is: ... the number of **cobalt-60** nuclei decay to one half of its initial value.

(b) Number of cobalt-60 atoms in 4.3×10^{-9} g, N

$$= \frac{4.3 \times 10^{-9} \times 10^{-3}}{60u}$$

$$= 4.317 \times 10^{13}$$

$$A = \lambda N$$

$$A = \left(\frac{\ln 2}{T_{\frac{1}{2}}} \right) N$$

$$T_{\frac{1}{2}} = \frac{N \ln 2}{A} = \frac{4.317 \times 10^{13} \times \ln 2}{1.8 \times 10^5}$$

$$T_{\frac{1}{2}} = 5.3 \text{ years}$$

5. **(a) (i)** The gravitational field strength g at a point is defined as the gravitational force per